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SUPERSEDING
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FEDERAL SPECIFICATION

DISK, ABRASIVE; MANDREL, DENTAL HANDPIECE

This Specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers brass-centered abrasive paper and plastic disks and mandrels suitable for polishing dental restorations.

1.2 Classification. The disks and mandrels covered by this specification shall be of the following types, grades, classes, and grit sizes as specified (see 6.1):

Type I - Disks.
Grade 1 - Paper.

Class 1 - Garnet.
Grit sizes - 120 and 180.
Class 2 - Silicon carbide.
Grit sizes - 100, 120, 150, and 320.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RSTS, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Grade 2 - Plastic.
Class 1 - Aluminum oxide.
Grit size - 280.
Class 2 - Zirconium silicate.
Grit size - 325

Type II - Mandrels.
Class 1 - Angle handpiece.
Class 2 - Straight handpiece.

2 APPLICABLE DOCUMENTS

2.1 Specificaitons, standards, and drawings. The following specifications, standards, and drawings of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications

L-P-377	- Plastic Sheet Strip, Polyester.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

Federal Standards

FED-STD- 66	- Steel, Chemical Composition and Hardenability.
FED-STD-376	- Preferred Metric Units For General Use By The Federal Government.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

(Copies of listed federal and military standards, specifications, Commercial Item Descriptions (CIDS), listed in the Department of Defense Index of Specifications and Standards (DoDISS), should be obtained from the DoD Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Copies of industry association documents should be obtained from the sponsor. Copies of all other listed documents should be obtained from the contracting activity or as directed by the contracting activity).

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(Federal Government activities may obtain copies of Federal Standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies).

Military Specifications

MIL-P-116	- Methods of Preservation.
MIL-L-10547	- Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

23087 - Disk, Abrasive & Mandrel, Dental Handpiece, Spring type.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bid or requests for proposal shall apply.

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

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U.S. Department of Health, Education, and Welfare
Food and Drug Administration:

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204).

The United States Pharmacopeial Convention, Inc.:

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

U.S Department of Commerce, National Bureau of Standards:

PS8 Product Standard for Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Officer, Washington, D. C. 20402. Price may be obtained from the Superintendent of Documents).

American Society For Testing and Materials (ASTM) Standards:

D 882 Standard Methods of Tests for Tensile Properties of Thin Plastic Sheeting.

. (Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 Materials.

3.1.1 Abrasive. The abrasive shall be of crushed materials, crystalline in form, and free from extraneous materials. The abrasives shall conform to the mineral name specified in table I. Grit sizes 100, 120, 150, and 180 shall conform to Product Standard PS 8 for screen grit sizes (grit sizes which are coarse enough to be measured and controlled by the use of sieves). Grit sizes 280, and 320 shall conform to PS 8 for sedimentation grit sizes (grit sizes which are so fine that they have to be measured and controlled by a sedimentation method based on Stokes Law). Grit size 325 zirconium silicate shall be tested as specified in 4.4.4.2 to determine grit size and a minimum of 98 percent shall pass through a 325 mesh sieve.

3.1.1.1 Silicon carbide. Abrasive shall be a hard bluish-black crystalline mineral of silicon carbide.

3.1.1.2 Garnet. Abrasive shall be crushed from clean, hard, fast cutting garnet.

3.1.1.3 Zirconium silicate. Abrasive shall be clean zirconium silicate flour, free from all extraneous material.

3.1.2 Backing.

3.1.2.1 Grade 1, paper backing. Grade 1 backing shall be made of a flexible specialized technical paper, conforming to the weight designation specified in table I for the applicable disk size. Backing for waterproof silicon carbide disks (grit sizes 150 and 320) shall be so treated with a saturant which will render the disks waterproof and suitable for both wet and dry operations.

3.1.2.2 Grade 2, plastic backing. Grade 2, plastic backing shall be 0.0075 inch polyester film complying with the requirements of type I, general purpose film of L-P-377, except that the haze requirement shall not apply; the dimension requirements shall be in accordance with 3.2 of this specification. Tests for tensile strength and elongation shall be conducted in accordance with 4.4.5 of this specification.

3.1.3 Adhesive. Adhesives specified in 3.1.3.1 and 3.1.3.2, shall be of such characteristics which will minimize dislodging of the abrasive grain from the backing when tested as specified in 4.4.3.

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3.1.3.1 Adhesive for grade 1, paper backed disks. The adhesive shall be a good quality, animal hide glue base, suitable for the purpose intended, garnet disks (grit sizes 120 and 180), and silicon carbide disks (grit sizes 100 and 120). The adhesive shall be a good quality synthetic varnish for waterproof silicon carbide disks (grit sizes 150 and 320).

3.1.3.2 Adhesive for grade 2, plastic backed disks. The agent for bonding the abrasives to the polyester film shall be a high quality, waterproof resin type adhesive suitable for the purpose intended.

3.1.4 Brass. Brass center for disks shall be of a grade suitable for the purpose intended.

3.1.5 Steel. Mandrels shall be made of steel conforming in composition to B 1113 or C 1117 of FED-STD No. 66 and shall be case hardened.

3.2 Description.

3.2.1 Type I, Disks. Disks shall be round with clearly trimmed edges. The square shaped mounting hole in the disk shall consist of a one-piece brass eyelet. The disks shall force fit the snap-on type, split flexible, spring tempered mandrels classes 1 and 2. Disks shall conform to the dimensions, style and design shown on Defense Personnel Support Center (DPSC) Drawing 23087. The disks shall conform to the requirements specified in table I for the grit size specified (see 1.2). Grade 1 disks shall have a paper backing, and shall be furnished in garnet (class 2) in grit sizes 120 and 180; and silicon carbide (class 3) in grit sizes 100, 120, 150, and 320. Grade 2 disks shall have a plastic backing, and shall be furnished in aluminum oxide (class 1), in grit size 280 and zirconium silicate (class 2) in grit size 325.

3.2.1.1 Coating. The abrasive coating shall be of the "closed coat" type. One side of the backing shall be coated with abrasive grit applied in such a manner as to completely cover the backing surface and produce a sharp, cutting abrasive product. For paper-backed disks only the back face of the disk shall be smoothly covered with a minimum of two coats of shellac. Disks (back face) may be color coded at the manufacturer's option. When coloring materials are used in color coding disks, these coloring materials shall be in accordance with an official article in the Paramacopeia of the United States of America (U.S.P.) or National Formulary (N.F.), or a coal-tar color certified as harmless and suitable for color coding the disks under the terms of the Federal Food, Drug and Cosmetic Act.

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TABLE I. Characteristics

Commercial name	Grit size (mesh size)	Backing 1/	Diameter of Disks ± 1/64 inch
		PAPER BACKED DISKS	
Garnet	120	85-95	1/2 inch
Garnet	120	85-95	5/8 inch
Garnet	120	85-95	3/4 inch
Garnet	180	85-95	5/8 inch
Garnet	180	85-95	3/4 inch
Silicon carbide	100	125-135	3/4 inch
Silicon carbide	120	125-135	3/4 inch
Silicon carbide Waterproof	150	78-83	5/8 inch
Silicon carbide Waterproof	320	78-83	5/8 inch
		PLASTIC BACKED DISKS	
Aluminum oxide	280	.0075 inch	5/8 inch
Zirconium silicate	325	.0075 inch	5/8 inch

1/ The basis weight of the paper in paper-backed disks is calculated in pounds per ream of 480 sheets, 24 by 36 inches, when conditioned at 70° + 20° Fahrenheit (F) and 50 ± 2 percent relative humidity. Grit sizes 150 and 320 silicon carbide include paper and saturant. For plastic-backed disks, the listed value shall be the thickness of the polyester film.

3.2.2 Type II, Mandrels. Mandrels shall be of one-piece, split, flexible, spring-tempered construction so proportioned as to provide uniform expansion without failure, and shall be suitable for use with abrasive disks included in this specification. The force fit of the abrasive disks shall be effected without any distortion or fracture of the snap-on mandrel feature. Mandrels shall be ground true. Mandrels shall be made of steel as specified in 3.1.5 and shall have a bright nickel plating. The plating shall be smooth and adherent. Class 1 mandrels shall fit dental angle handpieces, including those utilizing a slinger, and class 2 mandrels shall fit dental straight handpieces. Mandrels shall conform to the dimensions, style, and design shown in DPSC Drawing 23087 for angle (Class 1) handpieces and straight (class 2) handpieces as specified.

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3.2.2.1 Corrosion resistance. The mandrels shall show no more than two percent corrosion, when boil tested as specified in 4.4.1.

3.2.2.2 Runout. The total indicated runout of the mandrels shall be less than 0.003 inch, when tested as specified in 4.4.2.

3.2.3 Unit of issue. Unless otherwise specified (see 6.1), a package containing a minimum of 50 disks of the grit size specified (see 1.2) of type I, grade 1 disks, or a package containing a minimum of 200 disks of the grit size specified (see 1.2) of type I, grade 2 disks, or a package containing 6 mandrels (type II, class 1 or 2) shall be furnished as the unit of issue.

3.2.4 Drawings. The drawings forming a part of this specification are not production drawings. Drawings establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the drawings to provide a product with proper design, configuration and function. No deviations from the dimensional limitations prescribed on the drawings are permissible without prior approval of the contracting officer.

3.3 Workmanship. All disks and mandrels shall be free from defects which detract from their appearance or impair their serviceability. Disks shall be flat, cleanly cut, and have the mandrel hole truly centered. All sharp edges of mandrels shall be broken.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II through IV (see 4.2.1 through 4.3.1) and 4.3.1.1.

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4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table II. Unit of product for examination purposes shall be unit of issue (50 disks for type I, grade 1 disks; 200 disks for type I, grade 2 disks; 1 mandrel for type II mandrels). Five disks shall be selected at random from each package (unit) sampled for examination of all defects except "count".

TABLE II. Inspection and acceptable quality levels

Type defect	Inspection level	AQL (percent defective)
Major A	S-4	1.0
Major B	S-1	2.5
Minor A	S-4	2.5
Minor B	S-1	6.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III, except that the acceptance number shall be zero for all sample sizes. For type II mandrels the unit of product for test purposes shall be one mandrel.

TABLE III. Tests

Characteristic	Requirement	Test Procedure	Inspection level
Corrosion test	3.2.2.1	4.4.1	S-2
Runout	3.2.2.2	4.4.2	S-2
Disk-grading grain size	3.1.1	4.4.4	1/
Disk backing	3.1.2.2	4.4.5	1/
Practical disk service test	3.1.3	4.4.3	S-2 2/

1/ Sufficient amount from each lot (control number) of the applicable component shall be taken as the sample unit.
2/ Five disks shall be selected at random from each package sampled.

4.3 Examination. The mandrels and disks shall be examined for defects, including but not limited to, those listed in table IV.

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TABLE IV. Classification of defects

Categories	Defects
Major A:	
101	Mandrel class not as specified.
102	Grit size on disk not as specified.
103	Foreign material present in disk abrasive.
104	Disk does not fit mandrel properly.
105	Disk abrasive coating not as specified.
106	Backing cracks present in disk.
107	Mandrel does not permit ready application of disk.
Major B:	
121	Mandrel shank diameter not as specified.
122	Mandrel latch end dimension not as specified (class 1).
123	Dimensions effecting force fit of disks on mandrels not as specified.
Minor A:	
201	Shank end not smoothly rounded.
202	Mandrels not deburred.
Minor B:	
221	Dimensions (other than 121, 122 and 123) not as specified.
222	Count not as specified.

4.4 Tests.

4.4.1 Boil test. Before testing, mandrels shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, the mandrels shall be boiled in distilled water in a glass beaker for 30 minutes and allowed to remain submerged for an additional 24 hours.

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4.4.2 Runout. Prior to slotting, the total indicated runout of the mandrel head shall be measured to the nearest 0.0005 inch at the .130/.128 diameter section adjacent to the head. As an alternate after slotting, runout measurements shall be taken at the solid portion remaining of the .130/.128 diameter section adjacent to the inner end of the slot. The mandrel shall be held in a vee-block or precision chuck and rotated slowly. The eccentricity of the mandrel shall be measured by a toolmaker's microscope or optical comparator. The difference between the maximum and minimum readings for one full rotation of the mandrel shall be taken as the total indicated runout.

4.4.3 Practical disk service test. At least one unit package of disks shall be subjected to a practical service test of short duration to determine whether the disks are suitable for the purpose intended. Water shall be used as a lubricant during the testing of grit size 150 and 320 silicon carbide waterproof disks and grit sizes 280 aluminum oxide and 325 zirconium silicate plastic-backed disks.

4.4.4. Abrasive grain size.

4.4.4.1 Grading for grit size (aluminum oxide, silicon carbide, or garnet). Grading to determine the grit size, including recovery of abrasive grain from the coated abrasive product, preparation of grain for grading, and procedures shall be in accordance with PS8 for aluminum oxide, silicon carbide, or garnet.

4.4.4.2 Grading for grit size (zirconium silicate). The grading of the zirconium silicate abrasive grit shall be tested in accordance with the following method: Five 10-gram samples shall be taken to be tested. The samples shall be dried at 110 degrees Centigrade (^{o}C.) until they are brought to constant weight. The 10-gram sample shall then be placed on a 325 mesh, 8-inch diameter sieve and a Ro-Tap testing sieve shaker shall be used to screen the sample for a total of 1000 cycles. Five determinations shall be made on five different samples and the average of the five determinations shall be considered the result.

4.4.5 Disc backing (type I, grade 2 disks). The tensile strength and elongation of the disk (prior to coating) shall be tested in accordance with method A of ASTM D882. The specimen size shall be 1 inch wide by 5 inches long. Elongation rate shall be one hundred percent per minute. Separation between jaws shall be 2 inches. (Thickness of specimen shall be measured before coating).

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4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Disks and mandrels shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package.

5.1.1.3.1 Disks. Fifty type I, grade 1 disks or two hundred type I, grade 2 disks of one class and grit size only shall be packaged in accordance with MIL-P-116, method III in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.1.3.2 Mandrels. Six mandrels of one type and class only shall be wrapped in neutral tissue paper or other non-corrosive barrier material and packaged in accordance with MIL-P-116, method IC-1.

5.1.1.4 Intermediate package.

5.1.1.4.1 Disks. Twelve unit packages containing type 1, grade 1 or ten unit packages containing type 1, grade 2 disks of one class and grit size only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.1.4.2 Mandrels. Twelve unit packages containing mandrels of one type and class only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

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5.1.2 Level C.

5.1.2.1 Unit package. The preservation for fifty type 1, grade 1 or two hundred type 1, grade 2 disks of one class and grit only or six mandrels of one type and class shall conform to the requirements of MIL-STD-794 for level C.

5.1.2.2 Intermediate package. Twelve unit packages containing type 1, grade 1 or ten unit packages containing type 1, grade 2 disks of one class and grit only or twelve mandrels of one type and class shall be packaged in accordance with the requirements of MIL-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. One hundred thirty five intermediate packages (1620 units) of type I, grade 1 disks or forty intermediate packages (400 units) of type I, grade 2 disks of one class and size only or forty-eight intermediate packages (576 units) containing mandrels of one type and class only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. One hundred thirty-five intermediate packages (1620 units) of type I, grade 1 disks or forty intermediate packages (400 units) of Type I, grade 2 disks of one class and size only or forty-eight intermediate packages (576 units) containing mandrels of one type 2 load and conforming to PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. One hundred thirty five intermediate packages (1620 units) type I, grade 1 disks or forty intermediate packages (400 units) of type I, grade 2 disks of one class and size only or forty-eight intermediate packages (576 units) containing mandrels of one type and class only shall be packed in accordance with the requirements of MIL-STD-794 for level C.

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5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet or moremore. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 54 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Disks or mandrels packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Disks or mandrels packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Civil agencies. Unless otherwise specified, markings for shipment shall be as specified in FED-STD-123.

5.3.2 Military activities.

5.3.3 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

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6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, grade, class and grit size required (see 1.2).
(c) If unit of issue other than that specified (see 3.2.3).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-086-3211	DISK, ABRASIVE, Silicon Carbide, 100 Grit, 3/4 inch (19.05 mm) Diameter, 50s
6520-00-086-3213	DISK, ABRASIVE, Silicon Carbide, 120 Grit, 3/4 inch (19.05 mm) Diameter, 50s
6520-00-086-3214	DISK, ABRASIVE, Garnet, 180 Grit, 3/4 inch (19.05 mm) Diameter, 50s.
6520-00-086-3215	DISK, ABRASIVE, Garnet, 120 Grit, 1/2 inch (1.27 cm) Diameter, 50s.
6520-00-086-3216	DISK, ABRASIVE, Garnet, 180 Grit, 5/8 inch (15.87 mm) Diameter 50s.
6520-00-086-3217	DISK, ABRASIVE, Garnet, 120 Grit, 3/4 inch (19.05 mm) Diameter, 50s.
6520-00-086-3218	DISK, ABRASIVE, Garnet, 120 Grit, 5/8 inch (15.87 mm) Diameter, 50s.
6520-00-086-3221	DISK, ABRASIVE, Silicon Carbide, 150 Grit, Paper Backing, 5/8 inch (15.87 mm) Diameter, 50s.
6520-00-086-3222	DISK, ABRASIVE, Silicon Carbide, 320 Grit, Paper Backing, 5/8 inch (15.87 mm) Diameter, 50s.
6520-00-971-8220	MANDREL, DENTAL HANDPIECE, Angle Handpiece, Spring Type, 6s.
6520-00-971-8221	MANDREL, DENTAL HANDPIECE, Straight Handpiece, Spring Type 6s.
6520-01-004-9027	DISK, ABRASIVE, Aluminum Oxide, 280 Grit, Plastic, 5/8 inch (15.87 mm) Diameter, 200s.
6520-01-005-0612	DISK, ABRASIVE, Zirconium Silicate, 325 Grit, Plastic, 5/8 inch (15.87 mm) Diameter, 200s.

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6.3 Subject term (key word) listing:
Abrasive Disks
Dental Handpiece Mandrel
Garnet
Mandrels
Paper Backed Disks
Plastic Disk

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

Civil Agency Coordinating Activities:
VA-OSS
PHS
FDA-MPQAS

05466

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1984

Location: NCR Disk #8

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GG-D-419E	DISK ABRASIVE, DENTAL HANDPIECE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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